

Abhishek Nimalan

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EDUCATION

University of Toronto

Toronto, Ontario

Bachelor of Applied Science in Computer Engineering + PEY Co-op

Sep 2022 - Dec 2027 (Expected)

- Completing Minor in Artificial Intelligence Engineering
- **Relevant Coursework:** Operating Systems (C), Algorithms, Computer Networking (Python), Software Design (C++), Programming Fundamentals (C++ OOP), Digital Systems (Verilog), Computer Security (C, Python, SQL), Computer Organization (C, Assembly),
- **Awards:** University of Toronto Entrance Scholarship (\$7500), Edward S. Rogers Sr. Admission Scholarship (\$3500)

TECHNICAL SKILLS

Languages: Python, Go, JavaScript, C, C++, HTML, CSS, TypeScript, Bash, Java, Verilog HDL, Assembly (Nios II),

Developer Tools: Linux, Unix, Git, Gerrit, Github, [Docker](#), Kubernetes, Profilers, Virtual Machines

Certifications: Ericsson 5G, [AI & Automation](#), [Cloud Native](#), and [Security](#)

EXPERIENCE

Mozilla

May 2025 - August 2026

Software Engineer Intern

Toronto, Ontario

- Built cross-platform profiling infrastructure and automated CI pipelines (**Python**, **JavaScript**), leveraging native profilers (**Simpleperf**, **ETW/Xperf**, **Samplify**, and **perf**) to generate symbolicated CPU profiles from browser benchmarks and performance testing, enabling engineers to detect regressions via interactive [Chromium](#) and [Firefox](#) profiles ([Blog post](#)).
- Improved Mozilla's mobile browser performance testing by [extending Browsertime](#), the open-source JavaScript-based automated browser testing engine, with support for **Android's Simpleperf** CPU profiler and **Google's Perfetto** tracing framework, enabling the capture of hardware-level performance data directly from browser tests.
- Standardized **Python** symbolication pipelines across Firefox's entire performance testing infrastructure, transforming raw profiler output and debug symbols into human-readable profiles for 100+ daily tests with ~95% uptime ([Blog post](#))
- Developed build scripts and **YAML** configurations to introduce critical **Rust**, **JavaScript**, and **Android** tooling into Firefox CI.
- All open-source contributions, collaborations, and bug reports can be found on [GitHub](#), [Bugzilla](#), or [Phabricator](#)

Ericsson

May 2024 - Aug 2024

Software Integration Engineer Intern

Mississauga, Ontario

- Developed a full-stack, cloud-native application that automatically configures **20+** vDU servers for Ericsson 5G Cloud RAN, **reducing setup time by 90%**, and **minimizing on-site installation costs**
- Built a RESTful **Go** backend API to generate server configuration files (**YAML**, **JSON**) & retrieve SSL certs, download software packages, and leverage **Ansible** Playbooks, **Bash** scripts & **Python** tools to configure servers in **~10 minutes**
- Built an efficient WebSocket system to send updates about the server deployment progress to the frontend user interface to visualize
- Deployed the app as a **Docker** containerized microservice and integrated with the production **Kubernetes** cluster

Robotics for Space Exploration

Sep 2022 - May 2024

Software Subteam Developer

Toronto, Ontario

- [Programmed](#) a Depth Camera to detect nearby obstacles, leveraging **Python**, **YOLOv8**, and **Roboflow** [Documentation](#)
- Developed [scripts](#) that use LIDAR data to map obstacles, assisting with the navigation of a Rover [Documentation](#)
- Created **Python scripts** to compute velocity using acceleration data from an IMU sensor and compress real-time video streams
- Tested various networking equipment and sensors using **Linux**, **Bash**, and custom [scripts](#) (**XML**, **Python**)

PROJECTS

Wearable Device with Object Detection | *Python, OpenCV, YOLOv8, Redis Cloud, Git*

- [2nd Place Winner and Best Use of Redis at MLH NewHacks 2023 Hackathon](#)
- Built with a team a wearable, edge ML device that detects objects in its nearby surroundings and displays its approximate location
- Implemented object detection for real-time camera footage using **YOLOv8** and **OpenCV** (~1s inference time)
- Leveraged the **Redis Python client** and **Matplotlib** to store real-time results in a cloud database and visualize it

Embedded Hardware (FPGA/Microcontroller) Projects | *C++, C, Assembly (NIOS II), Verilog, Arduino*

- Snake on FPGA: An advanced snake clone for the DE1-SoC FPGA using **C**, **Assembly**, and **Python**
- Tetris on FPGA: A Tetris clone for the DE1-SoC FPGA using **Verilog**, **Quartus Prime**, and **ModelSim**

GIS Mapping Application | *C++, Glade, GTK, Git, Bash, Linux, HTML, CSS, OpenMP*

- Developed a high-performance, **AI-powered**, GIS mapping application for tourists using **C++**, **GTK/Glade**, **CSS** and **Git**
- Implemented optimized data structures and algorithms to improve graphics rendering, and map loading times (~10% reduction)
- Performed software debugging, testing, and profiling using **Bash** scripts, **Valgrind**, and **GDB**. Optimized pathfinding algorithms with multi-threading (**OpenMP**)